

Database schema

PostgreSQL, 21 application tables plus Alembic's `alembic_version` (current head `f3b7d1e8a4c5`). All primary keys are UUIDs generated by the application. The entity-relationship diagram is in `erd.md`; tenant isolation is described in `multi-tenancy.md`.

Conventions

- **Multi-tenant.** Every tenant-owned table has `company_id` (FK → `companies.id`). Row-Level Security is enabled on them with a `tenant_isolation` policy for `fddt_app` (`company_id = fddt_current_company_id()`), the transaction-local `app.current_company_id` and a `platform_all` policy for `fddt_platform`. No role has `BYPASSRLS`.
- **Insert-only tables** (the app roles are granted only `SELECT` + `INSERT`): `audit_log`, `case_risk_assessments`, `case_reports`, `risk_rules` (rule edits insert a new version).
- **JSON columns** (`jsonb`) hold check output and snapshots; their shapes are documented in the pipeline docs.
- **Bounding boxes** anywhere in JSON use one convention: `{page (1-based), x, y, width, height}` as 0-1 fractions of the page, origin top-left.
- **Enums** are native Postgres enums; their values are listed under each table and again in the index at the bottom.

Table	Tenancy	fddt_app	Purpose
<code>companies</code>	RLS (own row)	SELECT	One row per client company (tenant). <code>is_active = false</code> suspends it: its users can't sign in and existing tokens stop working.
<code>company_usage_stats</code>	tenant · RLS	SELECT, INSERT, UPDATE	Per-company, per-UTC-day usage counters (cases, documents, files, storage bytes) for the platform billing dashboard; bumped in the same transaction as each upload/case/report and reconciled nightly.
<code>users</code>	tenant · RLS	SELECT	Accounts (email + bcrypt-hashed password). Company users have a <code>company_id</code> and a ranked role (<code>user</code> / <code>reviewer_11</code> / <code>reviewer_12</code>); platform admins have <code>role = platform_admin</code> and no company (CHECK constraint).
<code>cases</code>	tenant · RLS	SELECT, INSERT, UPDATE	The unit of work: one submission of one or more documents. Carries workflow <code>status</code> , the owning reviewer tier <code>assigned_tier</code> and the latest <code>risk_tier</code> .
<code>documents</code>	tenant · RLS	SELECT, INSERT, UPDATE	One row per uploaded PDF. The original bytes live in Blob Storage (immutable); this row holds the SHA-256, the extraction results (<code>extracted_fields</code> , <code>ocr_text</code>) and <code>processing_status</code> .
<code>document_checks</code>	tenant · RLS	SELECT, INSERT, UPDATE	One row per (document, check type) — the latest run of each automated check (extraction stages, every forensic check, signature detection...), upserted so a redelivered task never duplicates it. <code>result</code> is the check's raw JSON output; the run history is in <code>audit_log</code> .
<code>document_page_hashes</code>	tenant · RLS	SELECT, INSERT, UPDATE	Perceptual hash (pHash) of every rendered PDF page, one per (document, page), used to find duplicate/near-duplicate pages within the same company.
<code>cross_document_findings</code>	tenant · RLS	SELECT, INSERT, UPDATE, DELETE	Case-level reconciliation results: a shared field (amount, date, issuer) that disagrees between documents in the same case.

Table	Tenancy	fddt_app	Purpose
<code>issuer_registry</code>	tenant · RLS	SELECT, INSERT, UPDATE	A company's known vendors/schools/etc. that extracted issuer names are fuzzy-matched against. Rows are deactivated, never deleted. A new company starts empty.
<code>risk_rules</code>	tenant · RLS	SELECT, INSERT	A company's configurable scoring rules. Rows are immutable and versioned: an edit inserts <code>version + 1</code> ; the current rule is the highest version. Seeded from <code>risk_rule_templates</code> when the company is created.
<code>risk_rule_templates</code>	platform-only	—	PLATFORM-LEVEL (no <code>company_id</code>): the default rule set copied into every NEW company's <code>risk_rules</code> . Edited by platform admins; edits never change existing companies. Only <code>fddt_platform</code> can access it.
<code>risk_settings</code>	tenant · RLS	SELECT, INSERT, UPDATE	One row per company holding its <code>medium</code> / <code>high</code> score thresholds and the metadata score cap (<code>metadata_score_cap</code> , default 40: the most points all <code>metadata.*</code> rules add together).
<code>case_risk_assessments</code>	tenant · RLS	SELECT, INSERT	One row per scoring run of a case (insert-only history; unique per (case, evidence fingerprint)): score, tier, frozen reason text and a snapshot of the exact rule versions and thresholds used.
<code>risk_scores</code>	tenant · RLS	SELECT	LEGACY. Created by the initial migration before the versioned engine existed; nothing in the application reads or writes it. <code>case_risk_assessments</code> replaced it.
<code>case_actions</code>	tenant · RLS	SELECT, INSERT	Reviewer-facing history of human actions on a case (approve / reject / escalate ...) with the actor's role at the time. Complements, and is also mirrored into, <code>audit_log</code> .
<code>case_reports</code>	tenant · RLS	SELECT, INSERT	History of generated per-case forensic PDF reports (insert-only). The PDF lives in Blob Storage under <code>companies/{company}/cases/{case}/reports/</code> ; this row stores its SHA-256, size and the assessment it reflects.
<code>audit_log</code>	tenant · RLS	SELECT, INSERT	Append-only system of record: every automated check result, human action and settings change is inserted here and never updated or deleted. <code>company_id</code> NULL = a platform-level event (company created, template edits, usage reconciliation, platform-admin access), visible only to platform admins; a company user's <i>user created/updated</i> goes into that company's log.
<code>signature_references</code>	tenant · RLS	SELECT, INSERT, UPDATE	A signature/stamp region drawn on a document, cropped and stored as a PNG, used as the reference for comparisons.
<code>signature_matches</code>	tenant · RLS	SELECT, INSERT, UPDATE	Result of comparing one reference against the signature region of another document in the same case (vision-model verdict or pixel-identical reuse verdict); one row per (reference, document, scope).
<code>bulk_upload_cases</code>	tenant · RLS	SELECT, INSERT, UPDATE	
<code>bulk_uploads</code>	tenant · RLS	SELECT, INSERT, UPDATE	

companies

One row per client company (tenant). `is_active = false` suspends it: its users can't sign in and existing tokens stop working.

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>name</code>	varchar(255)	no		unique
<code>is_active</code>	boolean	no	<code>true</code>	
<code>created_at</code>	timestampz	no	<code>now()</code>	
<code>updated_at</code>	timestampz	no	<code>now()</code>	
<code>max_file_size_mb</code>	integer	no		
<code>max_zip_size_mb</code>	integer	no		

- **Index** `companies_name_key` on (name) — unique
- **Check** `ck_companies_max_file_size_mb_positive`: `CHECK ((max_file_size_mb > 0))`
- **Check** `ck_companies_max_zip_size_mb_positive`: `CHECK ((max_zip_size_mb > 0))`
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app`: SELECT · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

company_usage_stats

Per-company, per-UTC-day usage counters (cases, documents, files, storage bytes) for the platform billing dashboard; bumped in the same transaction as each upload/case/report and reconciled nightly.

Column	Type	Null	Default	Notes
<code>company_id</code>	uuid	no		PK, FK → <code>companies.id</code>
<code>usage_date</code>	date	no		PK
<code>cases_created</code>	integer	no	<code>0</code>	
<code>documents_uploaded</code>	integer	no	<code>0</code>	
<code>files_stored</code>	integer	no	<code>0</code>	
<code>storage_bytes</code>	bigint	no	<code>'0'::bigint</code>	
<code>updated_at</code>	timestampz	no	<code>now()</code>	

- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

users

Accounts (email + bcrypt-hashed password). Company users have a `company_id` and a ranked role (`user` / `reviewer_11` / `reviewer_12`); platform admins have `role = platform_admin` and no company (CHECK constraint).

Column	Type	Null	Default	Notes
id	uuid	no		PK
created_at	timestamptz	no		
updated_at	timestamptz	no		
email	varchar(320)	no		unique
hashed_password	varchar(255)	no		
full_name	varchar(255)	yes		
role	enum user_role	no		
is_active	boolean	no		
company_id	uuid	yes		FK → companies.id

- **Index** ix_users_company_id on (company_id)
- **Index** ix_users_email on (email)
- **Index** users_email_key on (email) — unique
- **Check** ck_users_platform_admin_has_no_company: CHECK (((role = 'platform_admin'::user_role) = (company_id IS NULL)))
- **Enum** user_role: user, reviewer_l1, reviewer_l2, admin, platform_admin
- **Row-Level Security** on: platform_all (ALL to fddt_platform); tenant_isolation (ALL to fddt_app)
- **Grants** fddt_app: SELECT · fddt_platform: SELECT, INSERT, UPDATE, DELETE

cases

The unit of work: one submission of one or more documents. Carries workflow `status`, the owning reviewer tier `assigned_tier` and the latest `risk_tier`.

Column	Type	Null	Default	Notes
id	uuid	no		PK
created_at	timestamptz	no		
updated_at	timestamptz	no		
case_number	varchar(64)	no		unique
submitted_by_user_id	uuid	no		FK → users.id
status	enum case_status	no		
risk_tier	enum risk_tier	yes		
external_ref_id	varchar(255)	yes		
case_type	enum case_type	no		
assigned_tier	enum case_tier	no	'l1'::case_tier	
company_id	uuid	no		FK → companies.id
bulk_upload_id	uuid	yes		FK → bulk_uploads.id

Column	Type	Null	Default	Notes
reference_label	varchar(255)	yes		

- **Index** `cases_case_number_key` on (case_number) — unique
- **Index** `ix_cases_bulk_upload_id` on (bulk_upload_id)
- **Index** `ix_cases_case_number` on (case_number)
- **Index** `ix_cases_company_id` on (company_id)
- **Enum** `case_status`: `submitted`, `under_automated_review`, `pending_manual_review`, `auto_approved`, `escalated`, `under_investigation`, `approved`, `rejected`, `closed`
- **Enum** `risk_tier`: `low`, `medium`, `high`
- **Enum** `case_type`: `school_document`, `vendor_invoice`, `commercial_invoice`, `procurement_documentation`, `quotation`, `travel_reimbursement`, `other`
- **Enum** `case_tier`: `11`, `12`
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

documents

One row per uploaded PDF. The original bytes live in Blob Storage (immutable); this row holds the SHA-256, the extraction results (`extracted_fields` , `ocr_text`) and `processing_status` .

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>created_at</code>	timestampz	no		
<code>updated_at</code>	timestampz	no		
<code>case_id</code>	uuid	no		FK → <code>cases.id</code>
<code>uploaded_by_user_id</code>	uuid	no		FK → <code>users.id</code>
<code>original_filename</code>	varchar(512)	no		
<code>blob_storage_path</code>	varchar(1024)	no		
<code>file_hash</code>	varchar(64)	no		
<code>content_type</code>	varchar(255)	yes		
<code>file_size_bytes</code>	integer	yes		
<code>document_type</code>	varchar(128)	yes		
<code>processing_status</code>	enum <code>document_processing_status</code>	no		
<code>extracted_fields</code>	jsonb	yes		
<code>ocr_text</code>	text	yes		
<code>processing_error</code>	text	yes		
<code>company_id</code>	uuid	no		FK → <code>companies.id</code>

- **Index** `ix_documents_case_id` on (`case_id`)
- **Index** `ix_documents_company_id` on (`company_id`)
- **Index** `ix_documents_file_hash` on (`file_hash`)
- **Enum** `document_processing_status`: `pending`, `processing`, `complete`, `failed`
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

document_checks

One row per (document, check type) — the latest run of each automated check (extraction stages, every forensic check, signature detection...), upserted so a redelivered task never duplicates it. `result` is the check's raw JSON output; the run history is in `audit_log`.

Column	Type	Null	Default	Notes
<code>id</code>	<code>uuid</code>	no		PK
<code>created_at</code>	<code>timestampz</code>	no		
<code>updated_at</code>	<code>timestampz</code>	no		
<code>document_id</code>	<code>uuid</code>	no		FK → <code>documents.id</code>
<code>check_type</code>	enum <code>document_check_type</code>	no		
<code>status</code>	enum <code>document_check_status</code>	no		
<code>result</code>	<code>jsonb</code>	yes		
<code>confidence</code>	<code>double precision</code>	yes		
<code>error_message</code>	<code>varchar(2048)</code>	yes		
<code>company_id</code>	<code>uuid</code>	no		FK → <code>companies.id</code>

- **Index** `ix_document_checks_company_id` on (`company_id`)
- **Index** `ix_document_checks_document_id` on (`document_id`)
- **Index** `uq_document_checks_document_check_type` on (`document_id`, `check_type`) — unique
- **Enum** `document_check_type`: `ocr_layout`, `classification`, `field_extraction`, `issuer_verification`, `signature_stamp_detection`, `metadata_forensics`, `error_level_analysis`, `copy_move_detection`, `duplicate_detection`, `ai_content_detection`, `field_validation`, `visual_inconsistency_review`, `font_consistency`, `ghost_content`
- **Enum** `document_check_status`: `pending`, `running`, `completed`, `failed`
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app`: SELECT, INSERT, UPDATE · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

document_page_hashes

Perceptual hash (pHash) of every rendered PDF page, one per (document, page), used to find duplicate/near-duplicate pages within the same company.

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>created_at</code>	timestamptz	no		
<code>updated_at</code>	timestamptz	no		
<code>document_id</code>	uuid	no		FK → <code>documents.id</code>
<code>page_number</code>	integer	no		
<code>phash</code>	varchar(32)	no		
<code>company_id</code>	uuid	no		FK → <code>companies.id</code>

- **Index** `ix_document_page_hashes_company_id` on (`company_id`)
- **Index** `ix_document_page_hashes_document_id` on (`document_id`)
- **Index** `ix_document_page_hashes_phash` on (`phash`)
- **Index** `uq_document_page_hashes_document_page` on (`document_id`, `page_number`) — unique
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app` : SELECT, INSERT, UPDATE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

cross_document_findings

Case-level reconciliation results: a shared field (amount, date, issuer) that disagrees between documents in the same case.

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>created_at</code>	timestamptz	no		
<code>updated_at</code>	timestamptz	no		
<code>case_id</code>	uuid	no		FK → <code>cases.id</code>
<code>field_name</code>	varchar(128)	no		
<code>finding_type</code>	varchar(128)	no		
<code>severity</code>	enum <code>finding_severity</code>	no		
<code>description</code>	varchar(2048)	no		
<code>document_ids</code>	jsonb	yes		
<code>company_id</code>	uuid	no		FK → <code>companies.id</code>

- **Index** `ix_cross_document_findings_case_id` on (`case_id`)
- **Index** `ix_cross_document_findings_company_id` on (`company_id`)
- **Enum** `finding_severity`: `info`, `low`, `medium`, `high`
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app` : SELECT, INSERT, UPDATE, DELETE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

issuer_registry

A company's known vendors/schools/etc. that extracted issuer names are fuzzy-matched against. Rows are deactivated, never deleted. A new company starts empty.

Column	Type	Null	Default	Notes
id	uuid	no		PK
created_at	timestamptz	no		
updated_at	timestamptz	no		
name	varchar(255)	no		
tax_id	varchar(64)	yes		
type	enum issuer_registry_type	no		
name_arabic	varchar(255)	yes		
is_active	boolean	no	true	
company_id	uuid	no		FK → companies.id
country	varchar(2)	yes		

- **Index** ix_issuer_registry_company_id on (company_id)
- **Index** ix_issuer_registry_name on (name)
- **Index** ix_issuer_registry_name_arabic on (name_arabic)
- **Enum** issuer_registry_type : vendor , school , government , other
- **Row-Level Security** on: platform_all (ALL to fddt_platform); tenant_isolation (ALL to fddt_app)
- **Grants** fddt_app : SELECT, INSERT, UPDATE · fddt_platform : SELECT, INSERT, UPDATE, DELETE

risk_rules

A company's configurable scoring rules. Rows are immutable and versioned: an edit inserts version + 1 ; the current rule is the highest version. Seeded from risk_rule_templates when the company is created.

Column	Type	Null	Default	Notes
id	uuid	no		PK
created_at	timestamptz	no		
updated_at	timestamptz	no		
rule_id	varchar(64)	no		
category	varchar(128)	no		
condition	jsonb	no		
weight	double precision	no		
severity	varchar(32)	no		
reason_template	varchar(1024)	no		

Column	Type	Null	Default	Notes
is_active	boolean	no		
check_type	varchar(64)	no		
version	integer	no		
effective_from	timestamptz	no		
updated_by	uuid	yes		FK → users.id
change_note	varchar(1024)	yes		
company_id	uuid	no		FK → companies.id

- **Index** ix_risk_rules_company_id on (company_id)
- **Index** ix_risk_rules_rule_id on (rule_id)
- **Index** uq_risk_rules_company_rule_id_version on (company_id, rule_id, version) — unique
- **Row-Level Security** on: platform_all (ALL to fddt_platform); tenant_isolation (ALL to fddt_app)
- **Grants** fddt_app : SELECT, INSERT · fddt_platform : SELECT, INSERT

risk_rule_templates

PLATFORM-LEVEL (no company_id): the default rule set copied into every NEW company's risk_rules. Edited by platform admins; edits never change existing companies. Only fddt_platform can access it.

Column	Type	Null	Default	Notes
id	uuid	no		PK
rule_id	varchar(64)	no		unique
category	varchar(128)	no		
check_type	varchar(64)	no		
condition	jsonb	no		
weight	double precision	no		
severity	varchar(32)	no		
reason_template	varchar(1024)	no		
is_active	boolean	no	true	
updated_by	uuid	yes		FK → users.id
created_at	timestamptz	no	now()	
updated_at	timestamptz	no	now()	

- **Index** risk_rule_templates_rule_id_key on (rule_id) — unique
- **Grants** fddt_app : — · fddt_platform : SELECT, INSERT, UPDATE

risk_settings

One row per company holding its `medium` / `high` score thresholds and the metadata score cap (`metadata_score_cap` , default 40: the most points all `metadata.*` rules add together).

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>created_at</code>	timestampz	no		
<code>updated_at</code>	timestampz	no		
<code>medium_threshold</code>	integer	no		
<code>high_threshold</code>	integer	no		
<code>updated_by</code>	uuid	yes		FK → <code>users.id</code>
<code>company_id</code>	uuid	no		FK → <code>companies.id</code> , unique
<code>metadata_score_cap</code>	integer	no	40	

- **Index** `ix_risk_settings_company_id` on (company_id)
- **Index** `uq_risk_settings_company_id` on (company_id) — unique
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app` : SELECT, INSERT, UPDATE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

case_risk_assessments

One row per scoring run of a case (insert-only history; unique per (case, evidence fingerprint)): score, tier, frozen reason text and a snapshot of the exact rule versions and thresholds used.

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>case_id</code>	uuid	no		FK → <code>cases.id</code>
<code>raw_score</code>	double precision	no		
<code>score</code>	integer	no		
<code>tier</code>	enum <code>risk_tier</code>	no		
<code>triggered_reasons</code>	jsonb	no		
<code>risk_rules_version_snapshot</code>	jsonb	no		
<code>evidence_fingerprint</code>	varchar(64)	no		
<code>computed_at</code>	timestampz	no		
<code>company_id</code>	uuid	no		FK → <code>companies.id</code>

- **Index** `ix_case_risk_assessments_case_id` on (case_id)
- **Index** `ix_case_risk_assessments_company_id` on (company_id)
- **Index** `ix_case_risk_assessments_computed_at` on (computed_at)

- **Index** `ix_case_risk_assessments_evidence_fingerprint` on (evidence_fingerprint)
- **Index** `uq_case_risk_assessments_case_evidence` on (case_id, evidence_fingerprint) — unique
- **Enum** `risk_tier`: low, medium, high
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT · `fddt_platform`: SELECT, INSERT

risk_scores

LEGACY. Created by the initial migration before the versioned engine existed; nothing in the application reads or writes it. `case_risk_assessments` replaced it.

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>created_at</code>	timestampz	no		
<code>updated_at</code>	timestampz	no		
<code>case_id</code>	uuid	no		FK → <code>cases.id</code>
<code>score</code>	integer	no		
<code>tier</code>	enum <code>risk_score_tier</code>	no		
<code>triggered_reasons</code>	jsonb	no		
<code>rules_snapshot</code>	jsonb	no		
<code>company_id</code>	uuid	no		FK → <code>companies.id</code>

- **Index** `ix_risk_scores_case_id` on (case_id)
- **Index** `ix_risk_scores_company_id` on (company_id)
- **Enum** `risk_score_tier`: low, medium, high
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

case_actions

Reviewer-facing history of human actions on a case (approve / reject / escalate ...) with the actor's role at the time. Complements, and is also mirrored into, `audit_log`.

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>created_at</code>	timestampz	no		
<code>updated_at</code>	timestampz	no		
<code>case_id</code>	uuid	no		FK → <code>cases.id</code>
<code>actor_user_id</code>	uuid	yes		FK → <code>users.id</code>
<code>action_type</code>	enum <code>case_action_type</code>	no		

Column	Type	Null	Default	Notes
notes	varchar(4096)	yes		
actor_role	varchar(32)	yes		
company_id	uuid	no		FK → companies.id

- **Index** `ix_case_actions_case_id` on (case_id)
- **Index** `ix_case_actions_company_id` on (company_id)
- **Enum** `case_action_type`: assign, comment, approve, reject, escalate, reopen
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT · `fddt_platform`: SELECT, INSERT, UPDATE, DELETE

case_reports

History of generated per-case forensic PDF reports (insert-only). The PDF lives in Blob Storage under `companies/{company}/cases/{case}/reports/`; this row stores its SHA-256, size and the assessment it reflects.

Column	Type	Null	Default	Notes
id	uuid	no		PK
case_id	uuid	no		FK → cases.id
generated_by_user_id	uuid	no		FK → users.id
generated_at	timestampz	no		
blob_url	text	no		
file_size_bytes	integer	no		
page_count	integer	no		
report_sha256	varchar(64)	no		
risk_assessment_id	uuid	yes		FK → case_risk_assessments.id
company_id	uuid	no		FK → companies.id

- **Index** `ix_case_reports_case_id` on (case_id)
- **Index** `ix_case_reports_company_id` on (company_id)
- **Index** `ix_case_reports_generated_at` on (generated_at)
- **Row-Level Security** on: `platform_all` (ALL to fddt_platform); `tenant_isolation` (ALL to fddt_app)
- **Grants** `fddt_app`: SELECT, INSERT · `fddt_platform`: SELECT, INSERT

audit_log

Append-only system of record: every automated check result, human action and settings change is inserted here and never updated or deleted. `company_id` NULL = a platform-level event (company created, template edits, usage reconciliation, platform-admin access), visible only to platform admins; a company user's *user created/updated* goes into that company's log.

Column	Type	Null	Default	Notes
id	uuid	no		PK
created_at	timestampz	no		
case_id	uuid	yes		FK → cases.id
document_id	uuid	yes		FK → documents.id
actor_user_id	uuid	yes		FK → users.id
event_type	varchar(128)	no		
event_data	jsonb	yes		
company_id	uuid	yes		FK → companies.id

- **Index** ix_audit_log_case_id on (case_id)
- **Index** ix_audit_log_company_id on (company_id)
- **Index** ix_audit_log_created_at on (created_at)
- **Index** ix_audit_log_document_id on (document_id)
- **Index** ix_audit_log_event_type on (event_type)
- **Row-Level Security** on: platform_all (ALL to fddt_platform); tenant_isolation (ALL to fddt_app)
- **Grants** fddt_app : SELECT, INSERT · fddt_platform : SELECT, INSERT

signature_references

A signature/stamp region drawn on a document, cropped and stored as a PNG, used as the reference for comparisons.

Column	Type	Null	Default	Notes
person_name	text	no		
signature_image_url	text	yes		
bounding_box	jsonb	yes		
source_document_id	uuid	no		FK → documents.id
source_case_id	uuid	no		FK → cases.id
created_by	uuid	no		FK → users.id
is_library	boolean	no	false	
id	uuid	no		PK
created_at	timestampz	no		
updated_at	timestampz	no		
company_id	uuid	no		FK → companies.id

- **Index** ix_signature_references_company_id on (company_id)
- **Index** ix_signature_references_source_case_id on (source_case_id)
- **Index** ix_signature_references_source_document_id on (source_document_id)
- **Row-Level Security** on: platform_all (ALL to fddt_platform); tenant_isolation (ALL to fddt_app)

- **Grants** `fddt_app` : SELECT, INSERT, UPDATE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

signature_matches

Result of comparing one reference against the signature region of another document in the same case (vision-model verdict or pixel-identical reuse verdict); one row per (reference, document, scope).

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>document_id</code>	uuid	no		FK → <code>documents.id</code>
<code>case_id</code>	uuid	no		FK → <code>cases.id</code>
<code>signature_reference_id</code>	uuid	no		FK → <code>signature_references.id</code>
<code>comparison_scope</code>	enum <code>comparison_scope</code>	no		
<code>result</code>	enum <code>signature_match_result</code>	no		
<code>reasoning</code>	text	yes		
<code>compared_at</code>	timestampz	no		
<code>company_id</code>	uuid	no		FK → <code>companies.id</code>

- **Index** `ix_signature_matches_case_id` on (`case_id`)
- **Index** `ix_signature_matches_company_id` on (`company_id`)
- **Index** `ix_signature_matches_document_id` on (`document_id`)
- **Index** `ix_signature_matches_signature_reference_id` on (`signature_reference_id`)
- **Index** `uq_signature_matches_ref_doc_scope` on (`signature_reference_id`, `document_id`, `comparison_scope`) — unique
- **Enum** `comparison_scope` : `in_case`, `library`
- **Enum** `signature_match_result` : `consistent`, `possibly_consistent`, `inconsistent`, `cannot_determine`, `identical_reuse`, `reused_different_signer`
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app` : SELECT, INSERT, UPDATE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

bulk_upload_cases

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>company_id</code>	uuid	no		FK → <code>companies.id</code>
<code>bulk_upload_id</code>	uuid	no		FK → <code>bulk_uploads.id</code>
<code>folder_index</code>	integer	no		
<code>folder</code>	varchar(1024)	no		
<code>status</code>	varchar(16)	no		
<code>error_code</code>	varchar(64)	yes		

Column	Type	Null	Default	Notes
<code>error_message</code>	text	yes		
<code>case_id</code>	uuid	yes		FK → <code>cases.id</code>
<code>case_number</code>	varchar(64)	yes		
<code>files</code>	jsonb	no		
<code>created_at</code>	timestampz	no	<code>now()</code>	
<code>updated_at</code>	timestampz	no	<code>now()</code>	

- **Index** `bulk_upload_cases_bulk_upload_id_folder_index_key` on (bulk_upload_id, folder_index) — unique
- **Index** `ix_bulk_upload_cases_bulk_upload_id` on (bulk_upload_id)
- **Index** `ix_bulk_upload_cases_company_id` on (company_id)
- **Row-Level Security** on: `platform_all` (ALL to `fddt_platform`); `tenant_isolation` (ALL to `fddt_app`)
- **Grants** `fddt_app` : SELECT, INSERT, UPDATE · `fddt_platform` : SELECT, INSERT, UPDATE, DELETE

bulk_uploads

Column	Type	Null	Default	Notes
<code>id</code>	uuid	no		PK
<code>company_id</code>	uuid	no		FK → <code>companies.id</code>
<code>uploaded_by_user_id</code>	uuid	no		FK → <code>users.id</code>
<code>original_filename</code>	varchar(512)	no		
<code>case_type</code>	enum <code>case_type</code>	no		
<code>blob_storage_path</code>	varchar(1024)	no		
<code>file_hash</code>	varchar(64)	no		
<code>zip_size_bytes</code>	bigint	no		
<code>status</code>	enum <code>bulk_upload_status</code>	no		
<code>error_code</code>	varchar(64)	yes		
<code>error_message</code>	text	yes		
<code>case_folder_count</code>	integer	yes		
<code>cases_created</code>	integer	no	<code>0</code>	
<code>cases_failed</code>	integer	no	<code>0</code>	
<code>documents_accepted</code>	integer	no	<code>0</code>	
<code>documents_rejected</code>	integer	no	<code>0</code>	
<code>zip_details</code>	jsonb	yes		
<code>started_at</code>	timestampz	yes		
<code>finished_at</code>	timestampz	yes		

Column	Type	Null	Default	Notes
created_at	timestamptz	no	now()	
updated_at	timestamptz	no	now()	

- **Index** ix_bulk_uploads_company_id on (company_id)
- **Index** ix_bulk_uploads_uploaded_by_user_id on (uploaded_by_user_id)
- **Enum** case_type: school_document, vendor_invoice, commercial_invoice, procurement_documentation, quotation, travel_reimbursement, other
- **Enum** bulk_upload_status: queued, ingesting, complete, failed
- **Row-Level Security** on: platform_all (ALL to fddt_platform); tenant_isolation (ALL to fddt_app)
- **Grants** fddt_app: SELECT, INSERT, UPDATE · fddt_platform: SELECT, INSERT, UPDATE, DELETE

Enum index

Enum	Values
bulk_upload_status	queued, ingesting, complete, failed
case_action_type	assign, comment, approve, reject, escalate, reopen
case_status	submitted, under_automated_review, pending_manual_review, auto_approved, escalated, under_investigation, approved, rejected, closed
case_tier	11, 12
case_type	school_document, vendor_invoice, commercial_invoice, procurement_documentation, quotation, travel_reimbursement, other
comparison_scope	in_case, library
document_check_status	pending, running, completed, failed
document_check_type	ocr_layout, classification, field_extraction, issuer_verification, signature_stamp_detection, metadata_forensics, error_level_analysis, copy_move_detection, duplicate_detection, ai_content_detection, field_validation, visual_inconsistency_review, font_consistency, ghost_content
document_processing_status	pending, processing, complete, failed
finding_severity	info, low, medium, high
issuer_registry_type	vendor, school, government, other

Enum	Values
<code>risk_score_tier</code>	<code>low</code> , <code>medium</code> , <code>high</code>
<code>risk_tier</code>	<code>low</code> , <code>medium</code> , <code>high</code>
<code>signature_match_result</code>	<code>consistent</code> , <code>possibly_consistent</code> , <code>inconsistent</code> , <code>cannot_determine</code> , <code>identical_reuse</code> , <code>reused_different_signer</code>
<code>user_role</code>	<code>user</code> , <code>reviewer_l1</code> , <code>reviewer_l2</code> , <code>admin</code> , <code>platform_admin</code>

Schema notes

- `user_role` **still contains** `admin`. The single-tenant `admin` role was migrated to `platform_admin`; Postgres cannot drop an enum value, so `admin` remains in the type but no row uses it and the application never assigns it.
- `cases.case_number` and `users.email` carry a `UNIQUE` constraint *plus* a separate non-unique index from the original migrations; uniqueness is enforced either way.
- `risk_scores` exists but nothing in the application uses it. It is safe to leave; dropping it needs a migration.
- **No `notifications` table.** Notifications are not part of the current build.
- **Enum values added by later migrations** use `ALTER TYPE ... ADD VALUE`; a downgrade leaves them in place.
- **New tenant-owned tables** need `TenantScopedMixin`, RLS with a `tenant_isolation` and a `platform_all` policy, and explicit grants in their migration — there are no default privileges, so a table without them is unreachable by the app.